Computer Network I

Reti di Calcolatori I

Università di Napoli Federico II – Scuola Politecnica e delle Scienze di Base
Corso di Laurea in Informatica

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Network Layer

Internet Protocol: IP Assignment

- Within a block of addresses, **IP addresses must be assigned** to individual interfaces.

- A **system administrator** can configure the IP addresses in two ways:
  - **Manually**: by assigning one-to-one IP addresses to hosts.
  - **Automatically**: the network autonomously assigns free IP to incoming hosts.

- The second approach (most common) can be done by using the **Dynamic Host Configuration Protocol (DHCP)**.

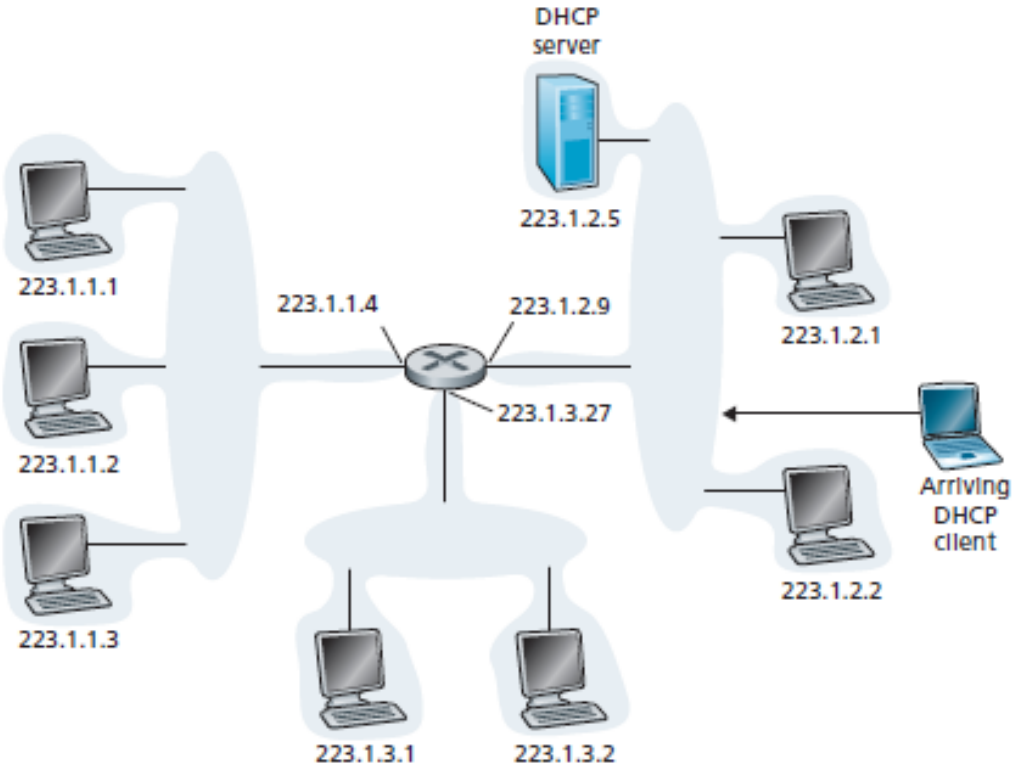
- In addition to host IP address, **DHCP provides a host with the information needed to join the network**, such as subnet mask, the address of the default gateway (to go outside of the network), and the address of the local DNS server.

- The **DHCP is plug-and-play**, and it is typically used in our homes.

Network Layer

Internet Protocol: DHCP

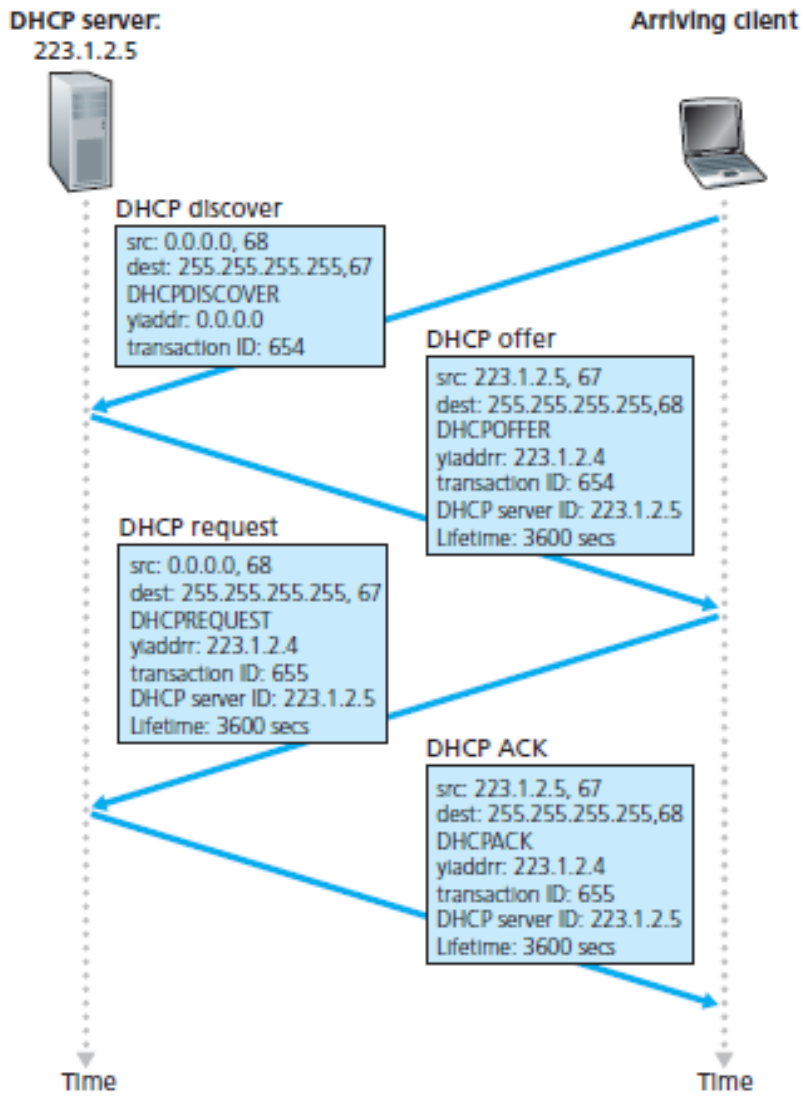
- The **DHCP is a client-server protocol**: a newly arriving host (client) connects to the DHCP server to receive network information**. DHCP service may be provided by a computer or by the router itself**.
  - Formally, **DHCP is an application-layer protocol**.
- Let's look to the previous example of **3 subnets connected by a single router**.
- Here we assume **the network on the right (233.1.2.0/24) to be endowed with a DHCP server**.
- When a new host joins the network, **it trades the IP address with the DHCP**.



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Internet Protocol: DHCP

- Adding a new host is mainly a 4-steps process:
  1. **DHCP server discovery**: the new host sends in **broadcast a DHCP discovery message (UDP on port 67)** to find the DHCP server.
     - Destination IP: 255.255.255.255 (broadcast).
     - Source IP: 0.0.0.0 (this host).
     - Random transaction ID.
  2. **DHCP server offer**: since multiple DHCP servers may be present, when a discovery message is received, a **server responds with a broadcast message (on port 68) offering a possible network configuration** (yiaddr filed – Your Internet ADDRess).
     - Destination IP: 255.255.255.255 (broadcast).
     - Source IP: 223.1.2.5 (DHCP server).
  3. **DHCP request**: the new client selects the accepted offer by **echoing back the offer message**.
  4. **DHCP ACK**: final **ACK message** confirming the transaction.



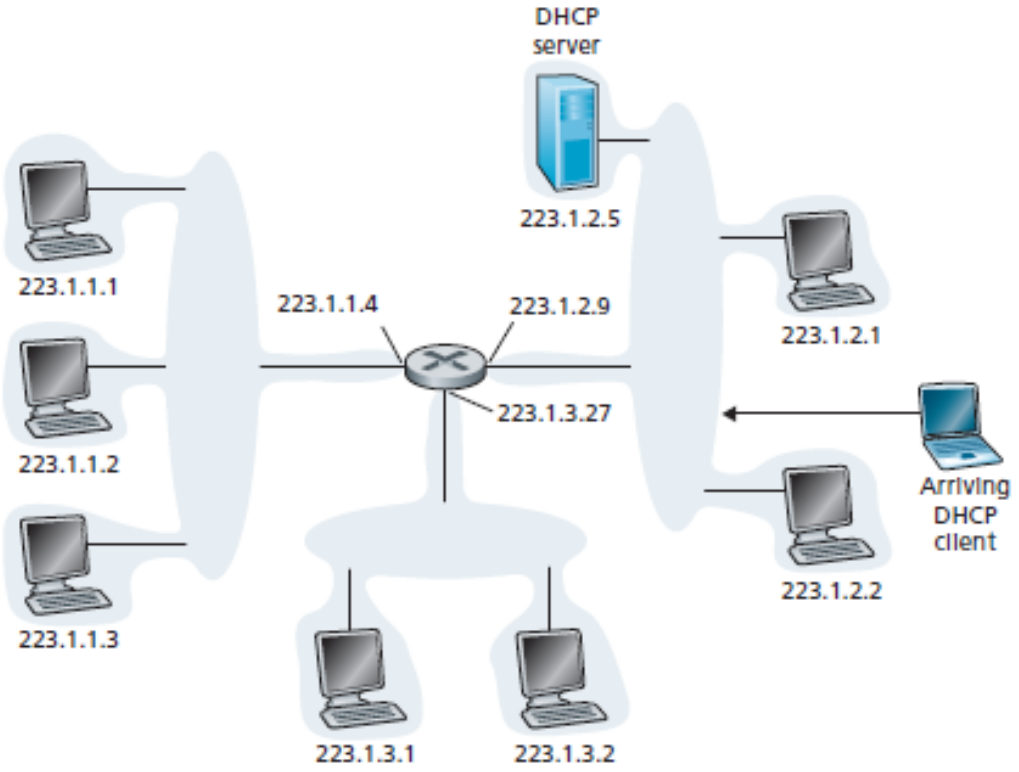
Network Layer

Internet Protocol: DHCP

- In this case everything works smoothly because the **DHCP and the arriving host are in the same subnet**.

- If a server exists but in a different subnet, we need a **DHCP relay agent** to forward DHCP messages (a router can do that).

- In this example we can configure our router to be a relay agent so that **also hosts from subnets 223.1.1.0/24 and 223.1.3.0/24 are served** by our DHCP.



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Route

- In Linux we can use the ifconfig in combination with the route command to see the network configuration provided by the DHCP.

- Usage:
  - See our current configuration (IP address, mask):
    - $ ifconfig
  - See our default gateway:
    - $ route -n

```
Lenovo-B50-80:~$ ifconfig wlp9s0
wlp9s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 100.103.0.64  netmask 255.255.0.0  broadcast 100.103.255.255
        inet6 fe80::47c4:ec38:2dc3:5d70  prefixlen 64  scopeid 0x20<link>
        ether 34:e6:ad:f0:d5:11  txqueuelen 1000  (Ethernet)
        RX packets 14398  bytes 20600768 (20.6 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4100  bytes 451906 (451.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

Lenovo-B50-80:~$ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         100.103.0.1     0.0.0.0         UG    600    0        0 wlp9s0
```

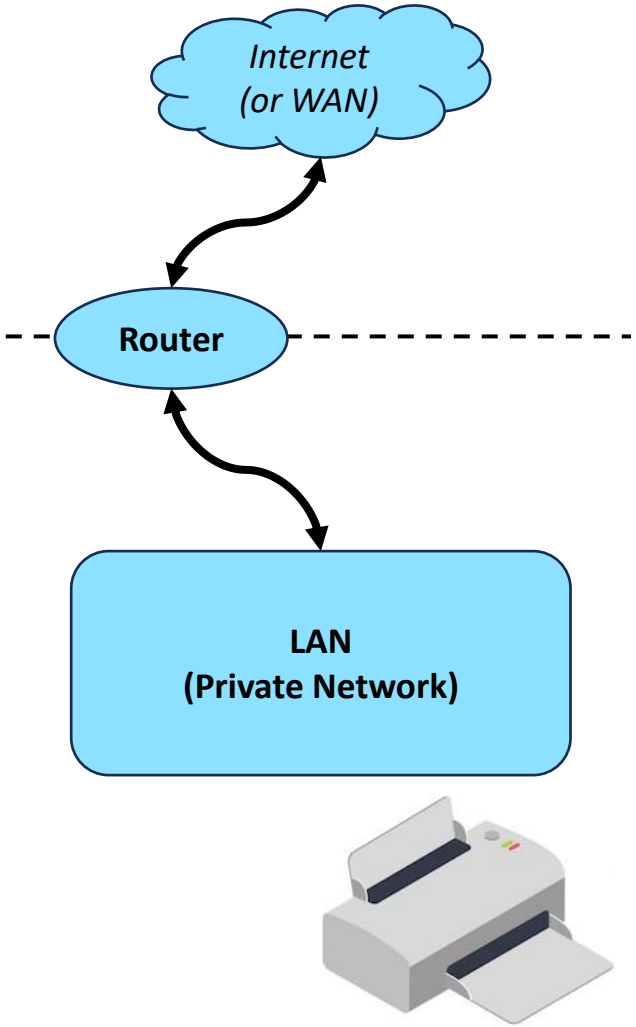
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Internet Protocol: Visibility

- In Internet there are billions of connected devices that must be **associated with unique IP addresses to be reached by anyone**.

- Is it really necessary **to have all devices visible to anyone**?
  - For example: is it good to have **my printer reachable (usable) by anyone** on Internet?

- There are **clear issues in having all devices reachable** from outside local networks:
  - IP **addresses are finite**.
  - If devices are locally connected **it is often unreasonable (or undesirable) to expose all of them on Internet**.
  - Local **administrators should be fully aware of the overlying IP block** structure in order to assign unused addresses (which is often not up to them, e.g., in home networks).



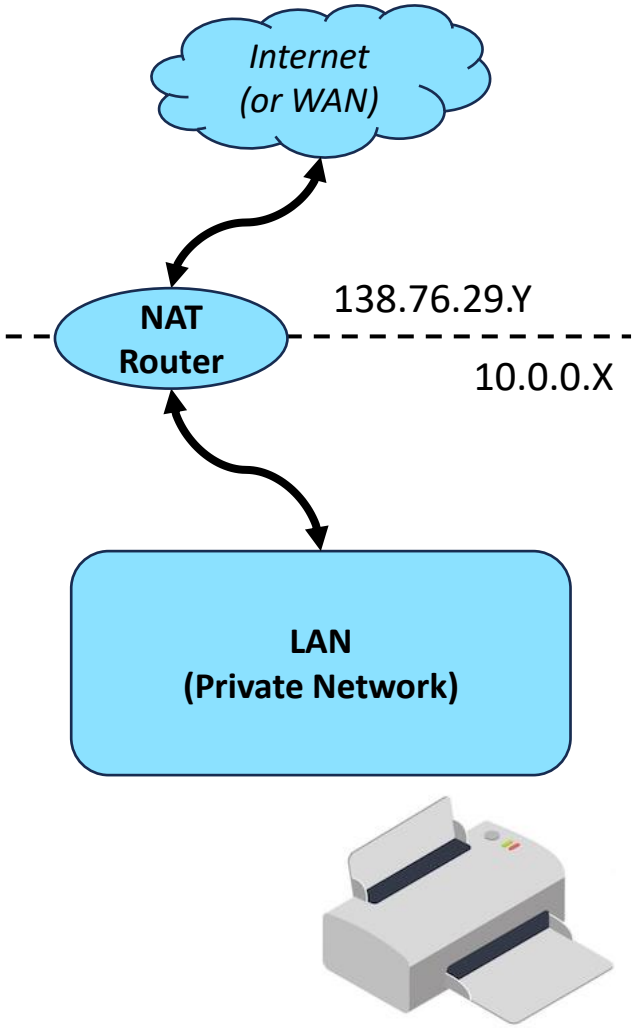
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Internet Protocol: NAT

- The **Network Address Translation (NAT)** service allows to **remap the IP addresses** of packets within a network into different ones (**network masquerading**).

- This is performed by using **a translation table** associating **multiple local IPs/ports (of the LAN) into a global IP/port (of the WAN)**. This service can be offered by **routers**.

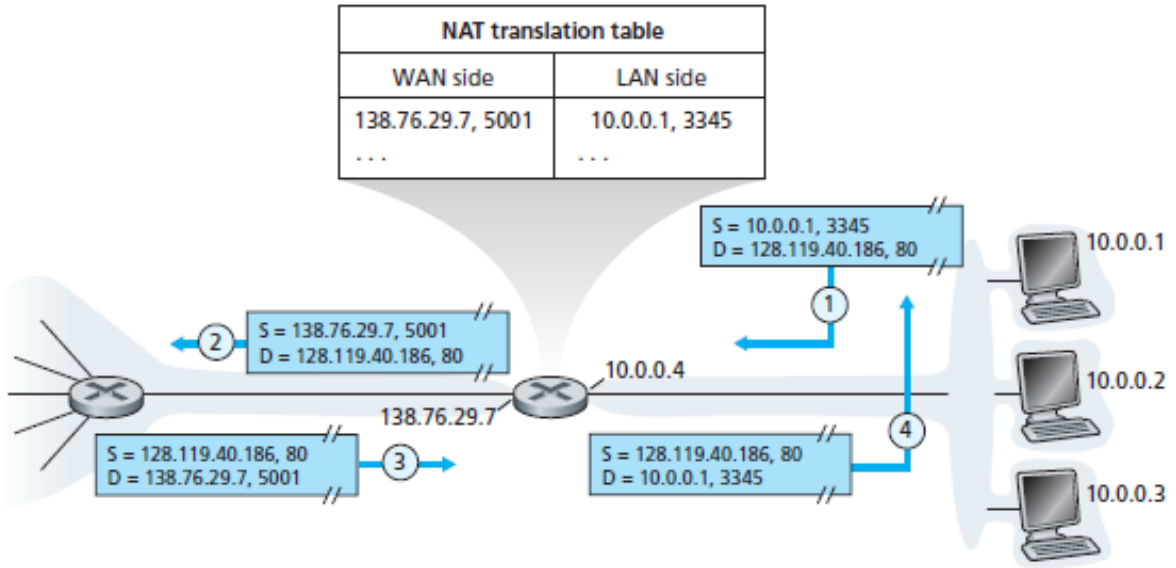
- The masqueraded local network is also called ***private network*** **or** ***realm with private addresses***. Private IP addresses are valid only inside the private network.



Network Layer

Internet Protocol: NAT

- This is an **example of a NAT-enabled router** connecting a network to internet.
- The four interfaces in the local network have the same subnet address of 10.0.0.0/24. The **router behaves as a message passer** that overrides the IPs with the one specified in the table:
  - **Outgoing packets** have their source IPs/ports overwritten **with a single WAN-side IP** (138.76.29.7) and a **fresh port** (5001).
  - **Incoming packets** have their destination IPs/ports overwritten **with the specific LAN-side IP** of the host (10.0.0.1) and the **initial port** (3345)
- It is important to notice that, since **hosts are unaware of the other hosts' traffic**, the **NAT cannot use the initial port** because multiple hosts may select the same port simultaneously.
- **Since the port is 16bits, NATs can manage over 60000 simultaneous connections**.



Network Layer

Ping

- To check if a host is reachable on a network, we can use the ping command (ping is the same on Linux/Windows machines).

- **Ping** (Packet Internet Groper) is a utility based on the **ICMP (Internet Control Message Protocol)** transport protocol and sends a “echo request” packet to a host that automatically answers with a “echo reply” message.
  - Ping also provide **the time elapsed between request/reply**, measuring the RTT.

- Usage:
  - To check the reachability:
    - $ ping [target address]

```
Lenovo-B50-80:~$ ping google.com
PING google.com (142.251.209.46) 56(84) bytes of data.
64 bytes from mil04s51-in-f14.1e100.net (142.251.209.46): icmp_seq=1 ttl=117 time=19.8 ms
64 bytes from mil04s51-in-f14.1e100.net (142.251.209.46): icmp_seq=2 ttl=117 time=20.3 ms
64 bytes from mil04s51-in-f14.1e100.net (142.251.209.46): icmp_seq=3 ttl=117 time=19.5 ms
64 bytes from mil04s51-in-f14.1e100.net (142.251.209.46): icmp_seq=4 ttl=117 time=18.9 ms
64 bytes from mil04s51-in-f14.1e100.net (142.251.209.46): icmp_seq=5 ttl=117 time=18.8 ms
64 bytes from mil04s51-in-f14.1e100.net (142.251.209.46): icmp_seq=6 ttl=117 time=29.1 ms
64 bytes from mil04s51-in-f14.1e100.net (142.251.209.46): icmp_seq=7 ttl=117 time=68.6 ms
64 bytes from mil04s51-in-f14.1e100.net (142.251.209.46): icmp_seq=8 ttl=117 time=19.6 ms
```

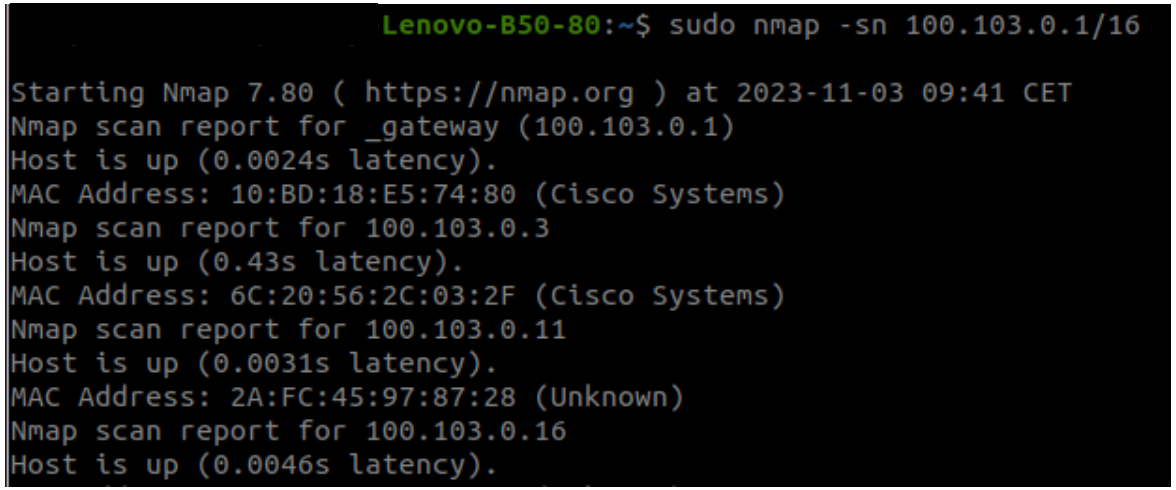
Network Layer

Nmap (pt.2)

- Besides port scanning, the nmap command can also be used to **map/scan the devices on the network**.

- It **relies on ping** (ICMP) to discover hosts on the network.

- Usage:
  - Ping-based hosts discovery:
    - $ sudo nmap -sn [gate address]/[subnet bits]



Network Layer

Internet Protocol: Reserved IP addresses

- Since **local administrators should be free to organize a private network** without interferences, by convention there are some **reserved IP blocks**, i.e., that ISPs/ICANN cannot assign to any organization. Some famous examples are:
  - 10.0.0.0 – 10.255.255.255 (reserved for **private networks**).
  - 192.168.0.0 – 192.168.255.255 (reserved for **private networks**).
  - 127.0.0.0 – 127.255.255.255 (indicate this machine, i.e., **loopback**).
    - Typically, 127.0.0.1 is used.
  - 0.0.0.0 (indicates **current network**).
    - Addresses having 0 in the host-part fall under this definition.
  - 255.255.255.255 (indicates **broadcast**).
    - Addresses having 255 in the host-part fall under this definition.
  - Etc.

- The necessity to reserve IPs is given by the fact that **private addresses may be the same as public (non-private) ones**, so routing within the network would be **ambiguous**.

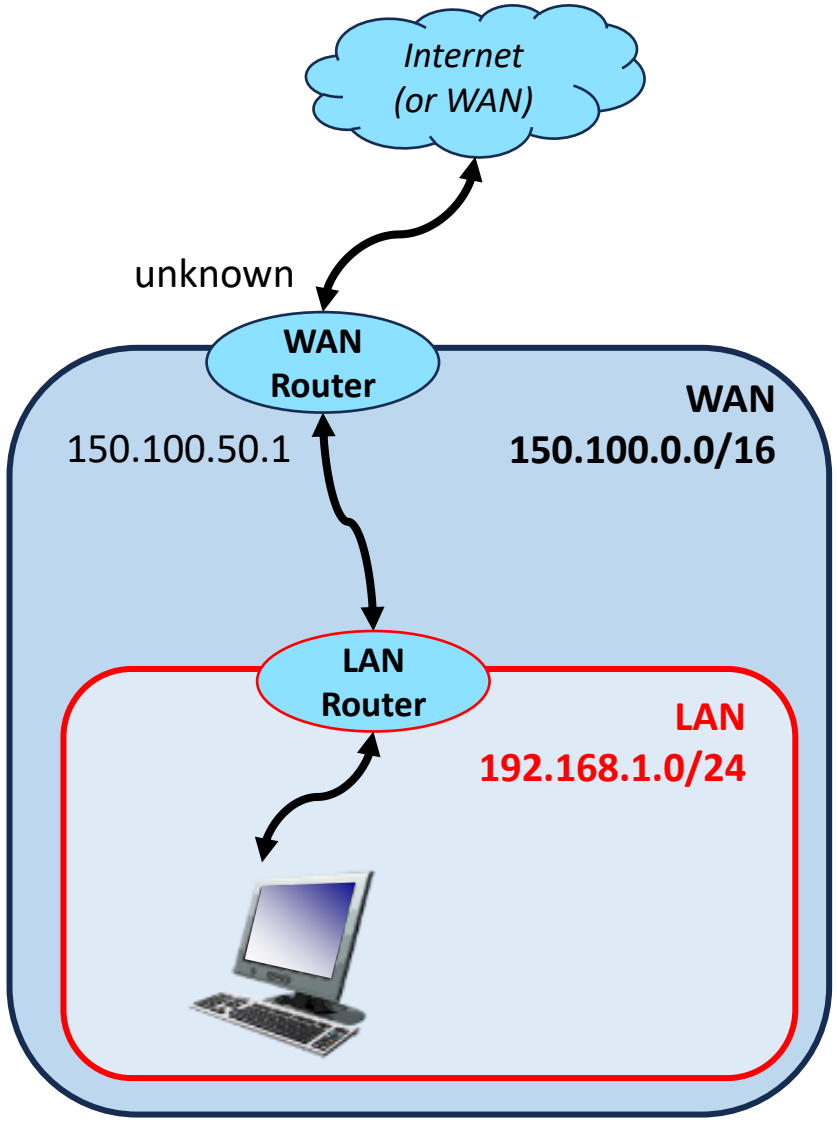
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Internet Protocol: LAN Setup Example

- Let's see and example in which we want to create **a new local subnetwork (LAN) into a pre-existing network (WAN)**.

- The **network administrator** of the WAN provides us the following information:
  - The **IP of the WAN**: 150.100.0.0/16.
  - The **IP of the getaway** (leading to external WAN/Internet): 150.100.50.1
  - A free **IP for our network**: 150.100.50.10

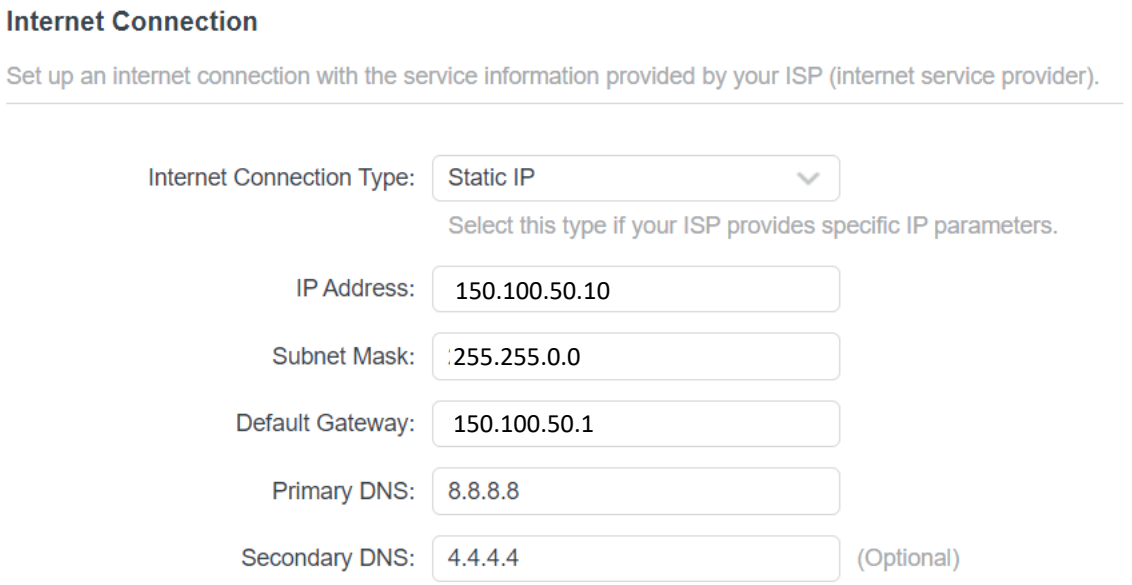
- We have now to configure our router and our devices to setup the new LAN.
  - We are considering a **NAT-enabled router**.
  - We decide that **the IP of the new LAN** will be: 192.168.1.0/24 (reserved for local networks).



Network Layer

Internet Protocol: LAN Setup Example

- Our **local router has 2 interfaces**, one for the WAN-side and another for the LAN-side.
- Let's start by configuring our local router's **WAN-side settings**.
- On the router we have to specify the **following parameters**:
  - The IP address of the router in the WAN.
  - The subnet of the WAN.
  - The gateway.
  - One or more DNS.
- Notice that we can also specify **a dynamic IP** if the WAN has a DHCP service.



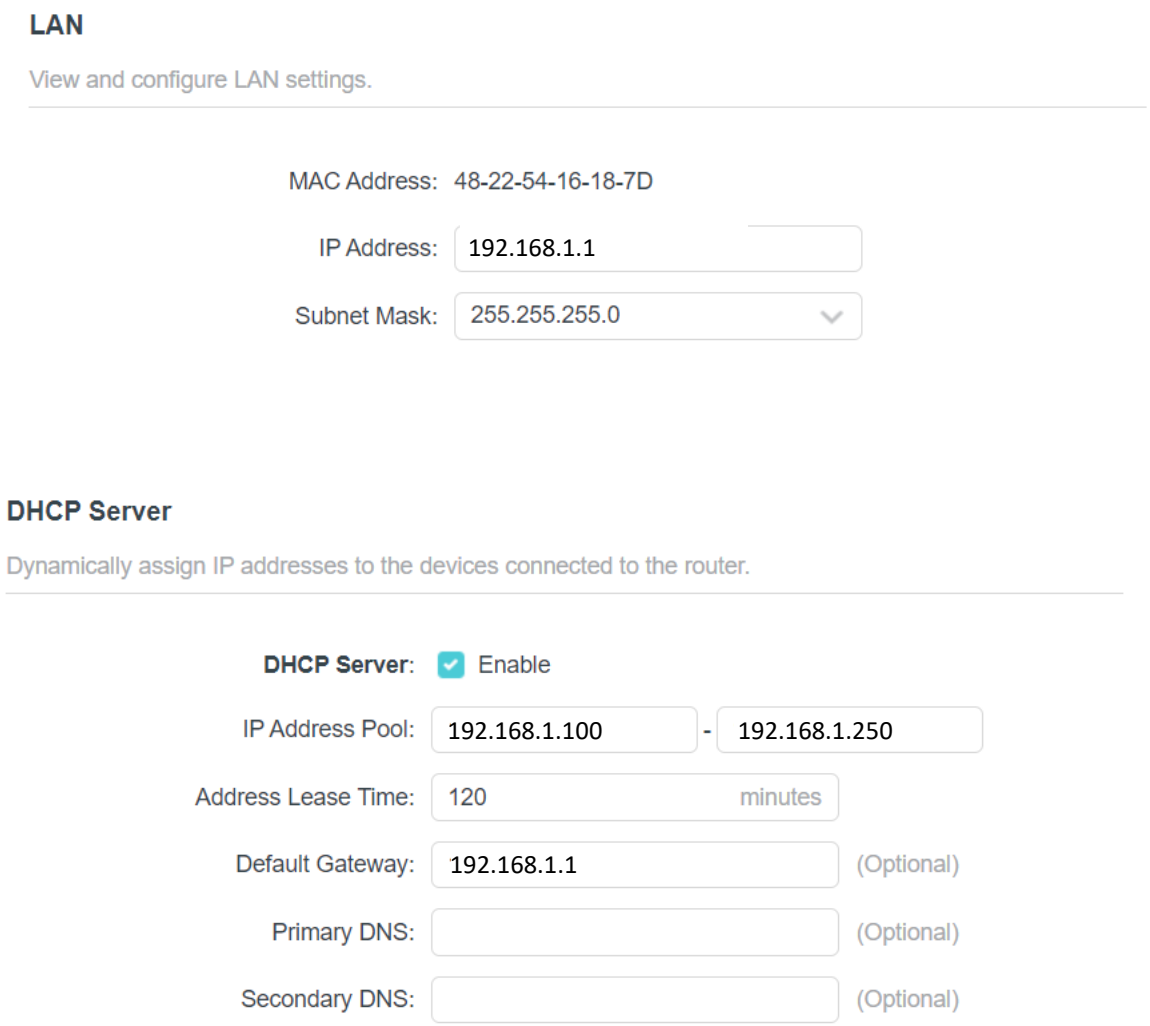
Network Layer

Internet Protocol: LAN Setup Example

- Now let's configure the **LAN-side settings**.

- We have to **give an IP** to the LAN-side interface of the router:
  - Since our network will be addressed as 192.168.1.0/24, we will give 192.168.1.1/24 to the router (it is typical to give .1 to it).

- In this example, our router also provide **DHCP service**, we can then specify:
  - **Address pool** (for DHCP hosts).
  - **Lease time** (timeout of the IP assignment, after that, the IP can be reused).
  - **Gateway** and **DNS** to be communicated to the hosts.

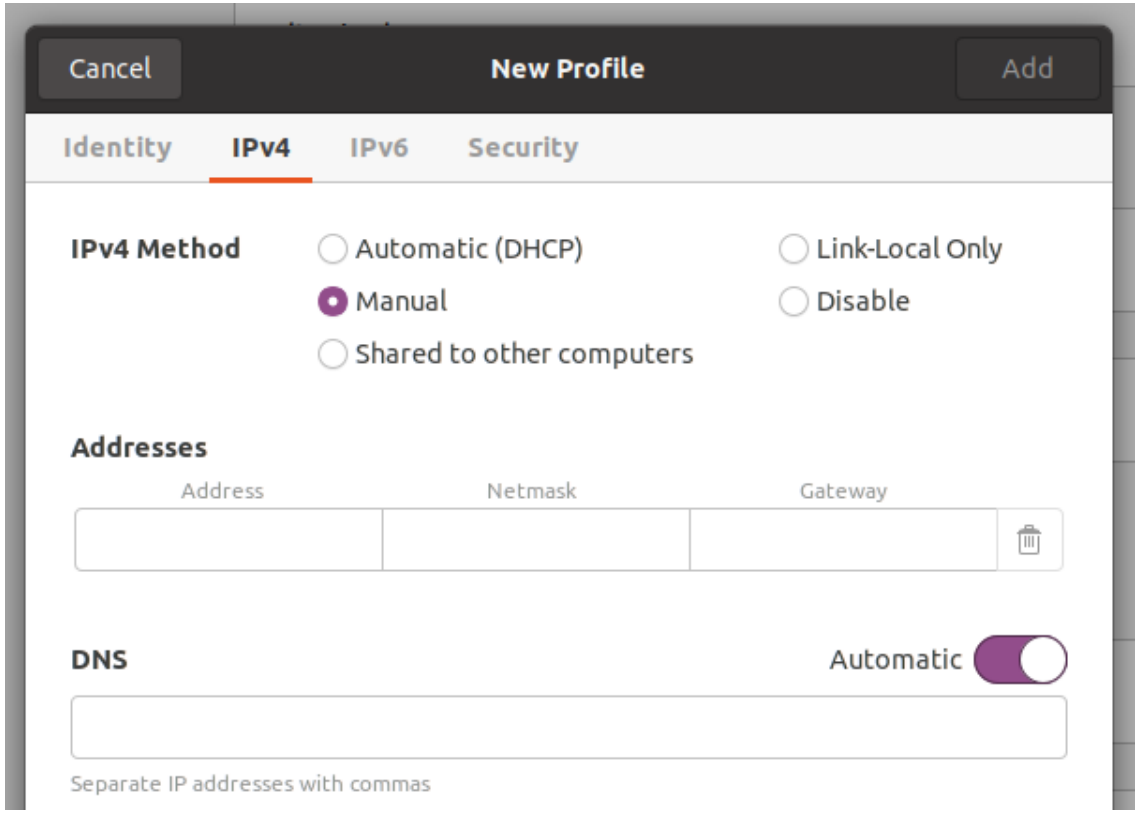


Network Layer

Internet Protocol: LAN Setup Example

- This is a simple example of how to configure this connection on a **new host** (Ubuntu 20.04).

- There are different methods to choose:
  - **Automatic (DHCP)**: uses DHCP to automatically configure the connection (you will get an IP from 192.168.1.100 to 192.168.1.250).
  - **Manual**: setup the connection by manually setting the configuration.

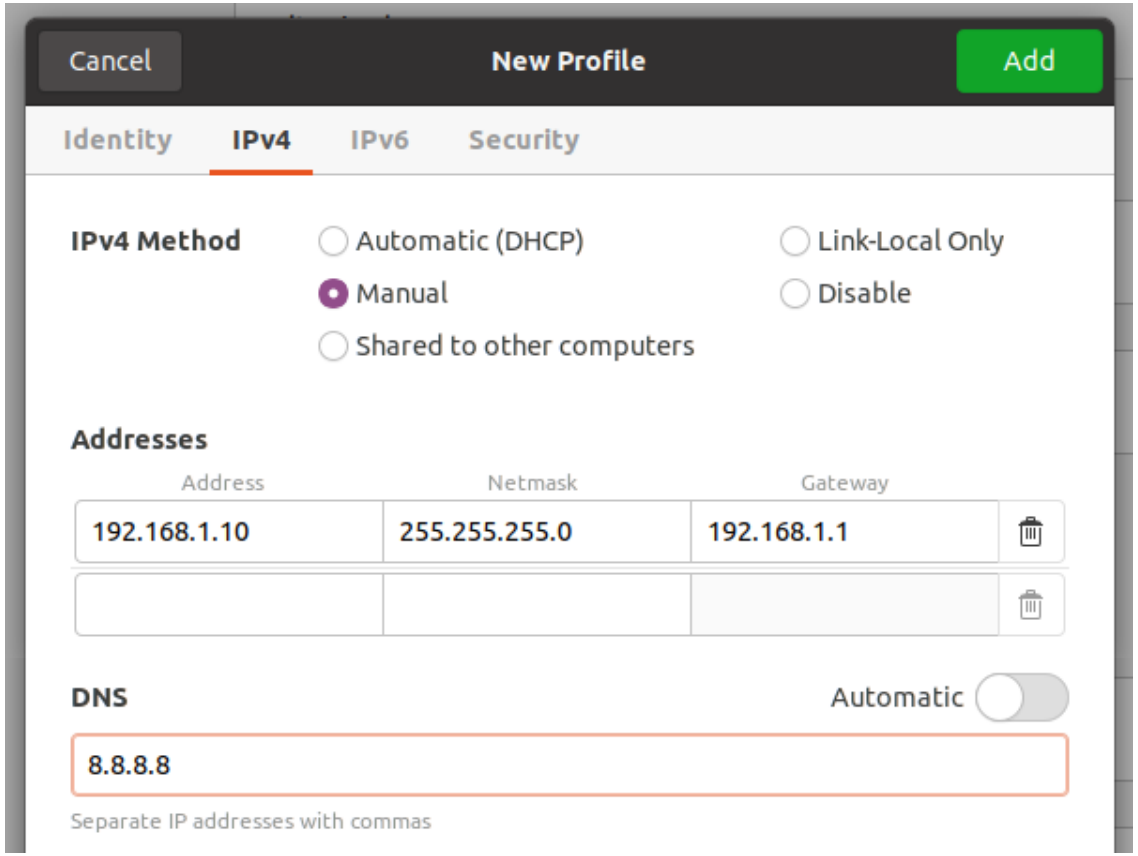


Network Layer

Internet Protocol: LAN Setup Example

- In a **manual setting** we need to know the network configuration (and the available IPs).

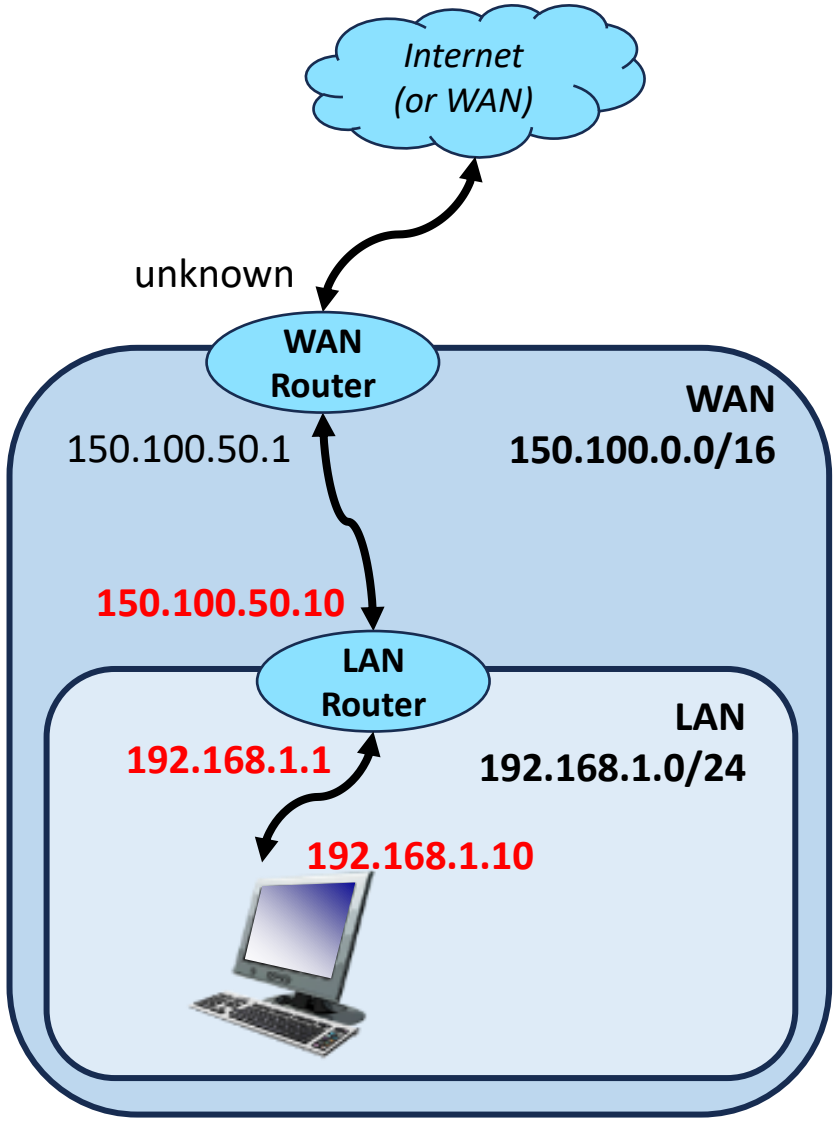
- We have to specify the main information for **connection setup**:
  - IP address of the host (static/fixed).
  - Subnet mask.
  - Gateway.
  - DNS (one or more).



Network Layer

Internet Protocol: LAN Setup Example

- In this created LAN we can **connect new hosts to the LAN** by specifying a manual IP address (outside the DHCP pool) or an automatic IP address (inside the DHCP pool).
- The connected **hosts will send their packets** to the local router (unaware of the outside WAN).
- We have the **local router configured to forward packets** to the WAN router (and possibly to the Internet).



Network Layer

Internet Protocol: IPv6

- In the early 1990s, the Internet Engineering Task Force (IETF), which is an ONG founded in 1986, **began an effort to develop a successor to the IPv4** protocol to face the 32-bit **IPv4 address shortage**.

- IPv6 was designed to **ensure more addresses available**, but it was also an opportunity to **update other aspects of IPv4**, based on the accumulated operational experience.

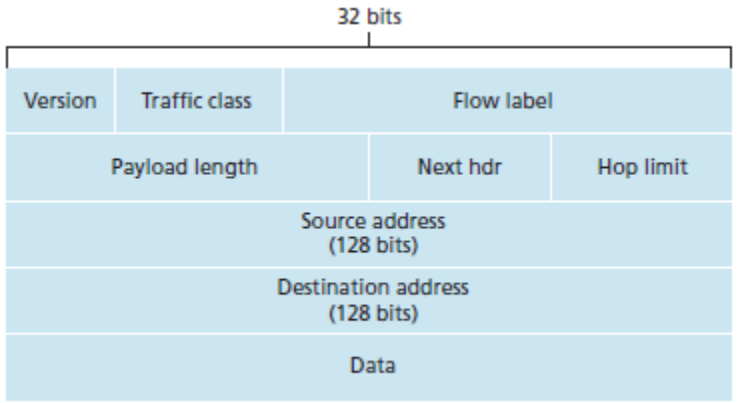
- It is **not sure when exactly all the available IPv4 would be exhausted** (it was speculated to be 2008 or 2018, but we have still some IPs left).

- But **what about IPv5?** It was proposed in 1979 and was based on the experimental Internet Stream Protocol (ST). It was still relying on 32bit addresses, so it was **dropped mainly because of addresses shortage**.

Network Layer

Internet Protocol: IPv6 Datagram

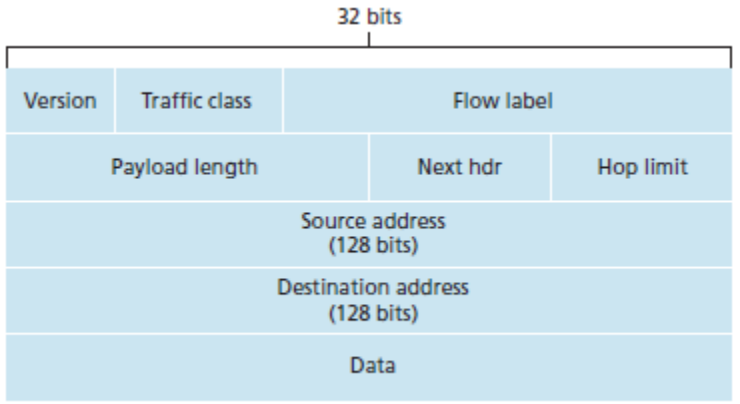
- IPv6 has the following **advantages**:
  - **Expanded addressing capabilities**: from 32 to 128 bits (i.e., 3.4028237e+38), which ensures that the world won't run out of IP addresses, as every grain of sand on the planet can be IP-addressable.
  - **A fixed-length 40-byte header**: some of IPv4 fields have been dropped or made optional.
  - **Flow labeling**: packets from some applications (e.g., audio/video streaming) can be grouped into a unique with flow having specific identification.
    - The precise role and usage of flows is still discussed.



Network Layer

Internet Protocol: IPv6 Datagram

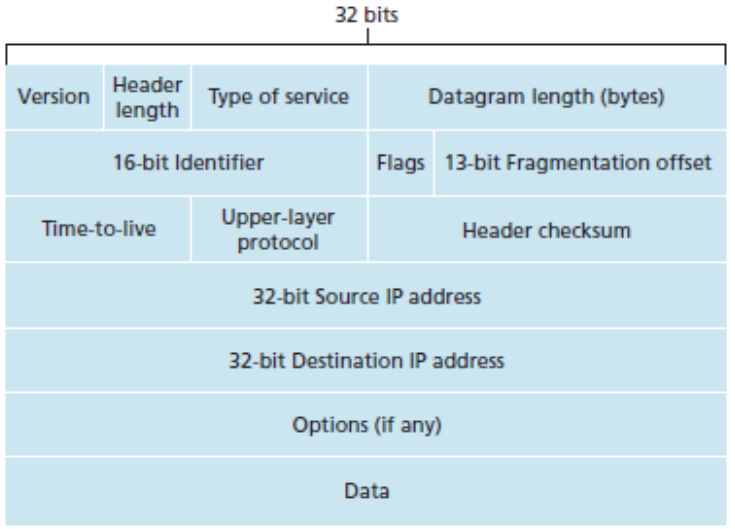
- IPv6 datagram is composed by the following fields:
  - **Version** (4 bits): identifies the **IP version** number. Not surprisingly, IPv6 carries a value of 6 in this field.
  - **Traffic class** (8 bits): like the TOS field in IPv4, can be used to give **priority to datagrams** (e.g., voice-over-IP over SMTP, etc.).
  - **Flow label** (20 bits): identify a **flow** of datagrams.
  - **Payload length** (16 bits): the **number of bytes of the payload** (40-bytes header excluded).
  - **Next header** (8 bits): identifies the **upper-level protocol** to which the contents (data field) of this datagram will be delivered (for example, to TCP or UDP). Similar to protocol field in the IPv4 header.
  - **Hop limit** (8 bits): Specifies the **time-to-live** as in TTL field of IPv4 (decremented every forward).
  - **Source and destination addresses** (128+128 bits): the IPv6 128-bit **addresses**.
  - **Data** (variable): the **payload** portion of the datagram.



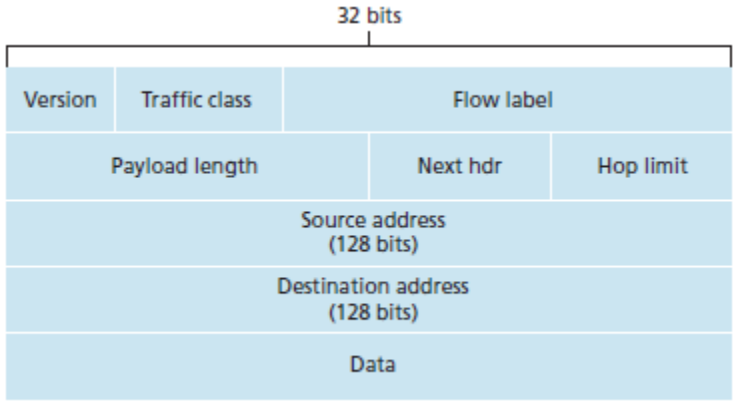
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Internet Protocol: IPv6 Datagram

- What is **missing** from IPv4:
  - **Fragmentation-related fields**: IPv6 **does not allow fragmentation and reassembly on intermediate routers**, but only on hosts.
    - If an IPv6 datagram received by a router is too large to be forwarded, the **router simply drops the datagram and sends a "Packet Too Big" error** message back to the sender. The sender can then resend the data, using a smaller IP datagram size.
  - **Header checksum**: it takes time on routers, and it is redundant as **link-layer protocols typically perform checksum** on the whole packet.
  - **Options**: are no longer a part of the standard IP header, **some options can be specified into next header** field (along with TCP/UDP identifiers).



IPv4



IPv6

Network Layer

Internet Protocol: From IPv4 to IPv6

- There is one big issue with IPv6: **it is not backward-compatible**. IPv4-capable systems are unable to handle IPv6 datagrams.

- How will Internet adopt IPv6?
  - **The flag day approach**: in a given time and date **all Internet machines would be turned off and upgraded from IPv4 to IPv6**. A similar approach have been used when TCP was introduced but it was a mess (even for the small internet network of the time). A flag day involving billions of devices is even more unthinkable today.
  - **The tunneling approach**: the use of specific routers (tunnels) in charge of **mapping IPv4 datagrams into IPv6 datagrams** and vice versa in almost-transparent way. This is probably the most realistic approach, already **used in practice**.

- While the adoption of IPv6 was initially slow, it is accelerating. Google reports that **about 45% of clients (in 2023) accessing Google** services use IPv6 (https://www.google.com/intl/en/ipv6/statistics.html).